WorkBench failure analysis: English vs Bangla (gemma4:31b and gpt-oss:20b)

Generated by `analysis/workbench/fa_report.py` on 2026-10-10, following `failure_analysis_pipeline.md` step by step. Every number below is recomputed from the paired run tables in `WorkBench-bangla-main/results/comparisons/`; every label comes from `labels_final.csv` (first annotator, blind second reviewer and adjudicator) in `results/failure_pipeline/`.

Summary of results

- **No detectable language gap for either model.** gemma4:31b passes 81.1% in English and 82.2% in Bangla (gap -1.1 pp, 95% interval [-10.0, 7.8] pp, McNemar p = 1.000); gpt-oss:20b passes 58.9% and 53.3% (gap +5.6 pp, interval [-5.6, 16.7] pp, p = 0.424). Both intervals include zero.
- **The pairs that flip go both ways.** gemma: 7 English-only passes against 8 Bangla-only passes; gpt-oss: 15 against 10.
- **Most Bangla-only failures are failures the model also makes in English.** Of the 22 EN-pass/BN-fail cases, 17 are coded general model weakness (gemma 3 of 7, gpt-oss 14 of 15).
- **Genuine Bangla failures are few and share one mechanism:** 3 cases (gemma 2, gpt-oss 1); the clearest are Bangla relative-date phrases ("গত সপ্তাহে", last week) grounded to a different date window, with correct tool syntax.
- **gpt-oss fails mainly on tool arguments in both languages**: wrong or missing filter values and invented argument names, independent of language.
- **Limits.** One run per cell at temperature 0; 90 tasks (15 per domain); all labels are model annotations awaiting native-speaker review of the translations.

---

0. Scope and how the pipeline was adapted

The pipeline was written for tau2-style dialogue benchmarks. This report applies it to **WorkBench, 90 pilot tasks (15 in each of six domains), two models (gemma4:31b and gpt-oss:20b), one English run (condition C0) and one Bangla run (condition C6) of each per task**, temperature 0. C6 = Bangla task and system prompt, English tool descriptions, replies forced to Bangla. Pass/fail is WorkBench's own check: the final database state must equal the ground truth, and every tool call must execute.

Pipeline item	In this report
Dataset	One: WorkBench. Dataset-wise distribution is replaced by domain (email, calendar, analytics, project management, CRM, multi-domain).
Model	Two models, each with a complete English and Bangla run on all 90 tasks.
Domain	The six WorkBench domains, 15 tasks each.
"Bangla user simulator"	Not applicable: WorkBench tasks are single-turn, with no simulated user. Confound C2 and decision-tree Step 2 are zero by construction.
"Dialogue-state failure"	Not applicable (single-turn). Kept in the taxonomy, zero cases.
Case selection (§3)	Wider than the pipeline asks: every EN-pass/BN-fail case, every EN-fail/BN-pass case (English side) and every EN-fail/BN-fail case (both sides) are labelled, so **every failed run has a code**.
Second reviewer	**Done.** A blind second reviewer labelled every EN-pass/BN-fail case; disagreements were adjudicated (Section 14).
Annotators	All three roles (annotator, blind reviewer, adjudicator) are Claude models. Native-speaker review of the translations is pending.

R. Results of all runs

Four runs are compared, each on all 90 tasks: gemma4:31b and gpt-oss:20b, each with an English and a Bangla prompt.

Run	Model	Prompt	Tasks	All	Email	Calendar	Analytics	Project mgmt	CRM	Multi-domain
GE	gemma4:31b	English	90	73 / 90 (81.1%)	15 / 15 (100.0%)	13 / 15 (86.7%)	13 / 15 (86.7%)	13 / 15 (86.7%)	11 / 15 (73.3%)	8 / 15 (53.3%)
GB	gemma4:31b	Bangla	90	74 / 90 (82.2%)	14 / 15 (93.3%)	14 / 15 (93.3%)	10 / 15 (66.7%)	14 / 15 (93.3%)	13 / 15 (86.7%)	9 / 15 (60.0%)
GO	gpt-oss:20b	English	90	53 / 90 (58.9%)	11 / 15 (73.3%)	10 / 15 (66.7%)	11 / 15 (73.3%)	8 / 15 (53.3%)	9 / 15 (60.0%)	4 / 15 (26.7%)
OB	gpt-oss:20b	Bangla	90	48 / 90 (53.3%)	10 / 15 (66.7%)	11 / 15 (73.3%)	7 / 15 (46.7%)	9 / 15 (60.0%)	7 / 15 (46.7%)	4 / 15 (26.7%)

R.1 The study in two figures

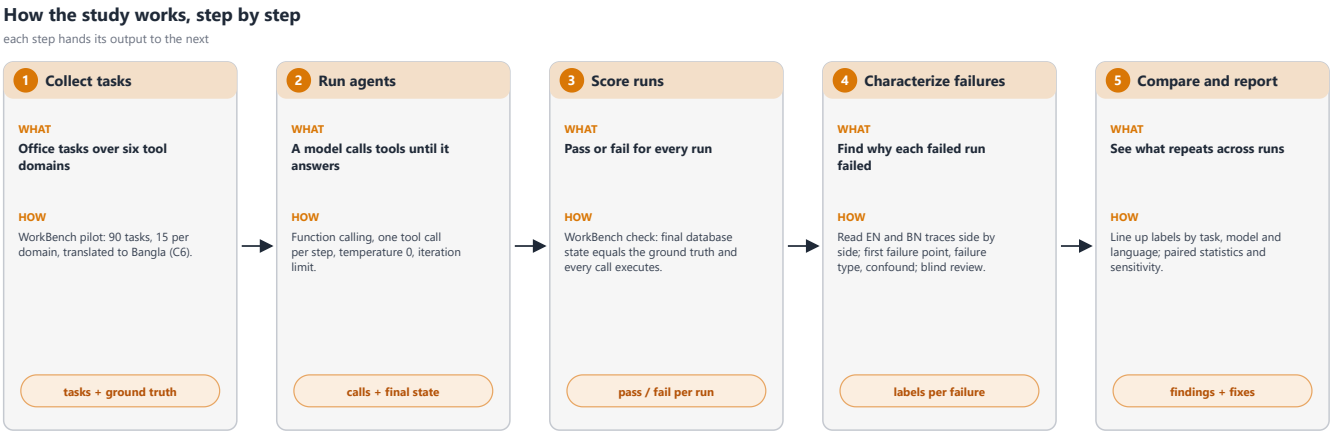


Figure 1. The pipeline, step by step: what each step does, how, and what it hands to the next.

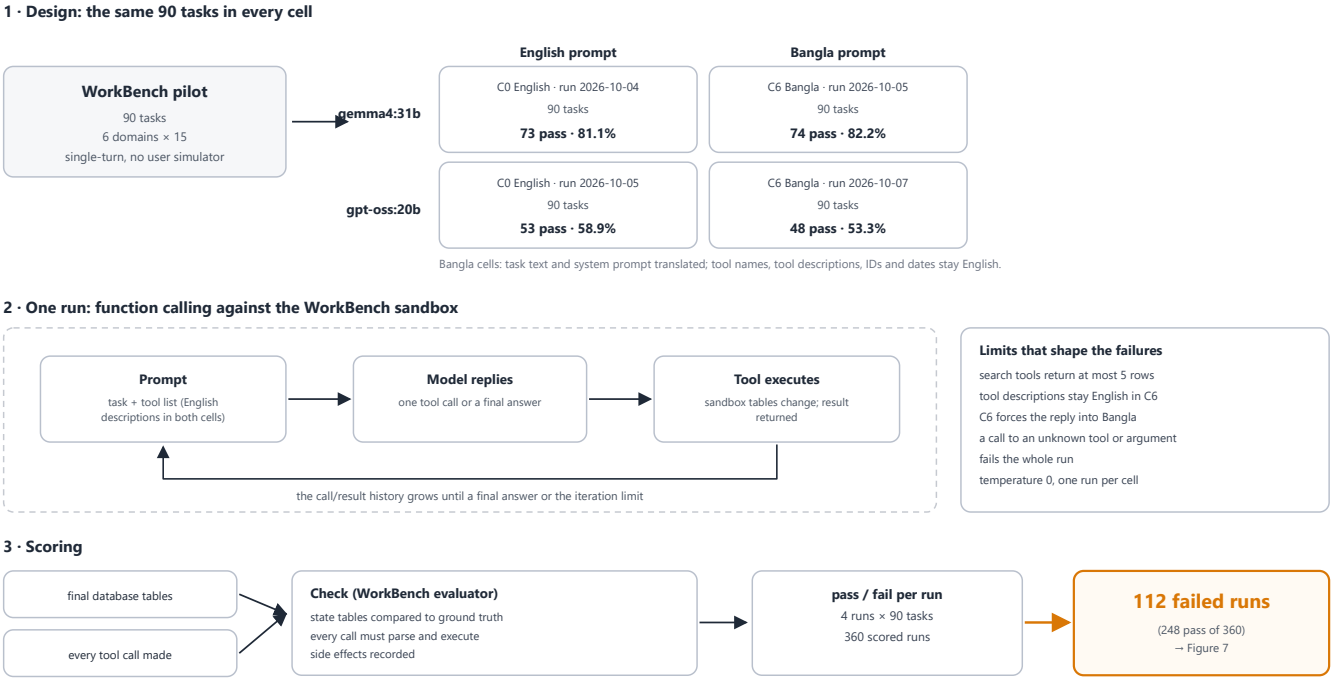


Figure 2. How the study works: the four cells, one run, and the scoring.

R.2 Success rate by run and domain

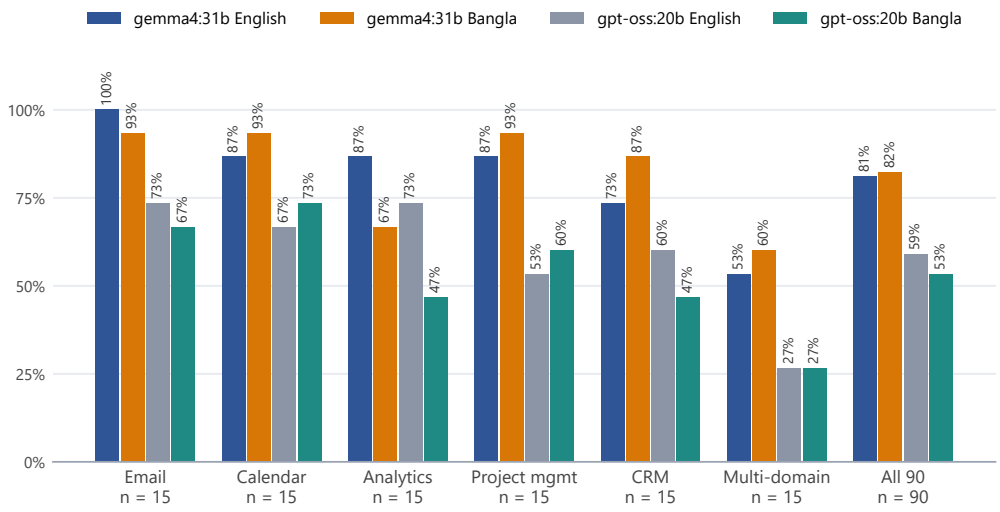
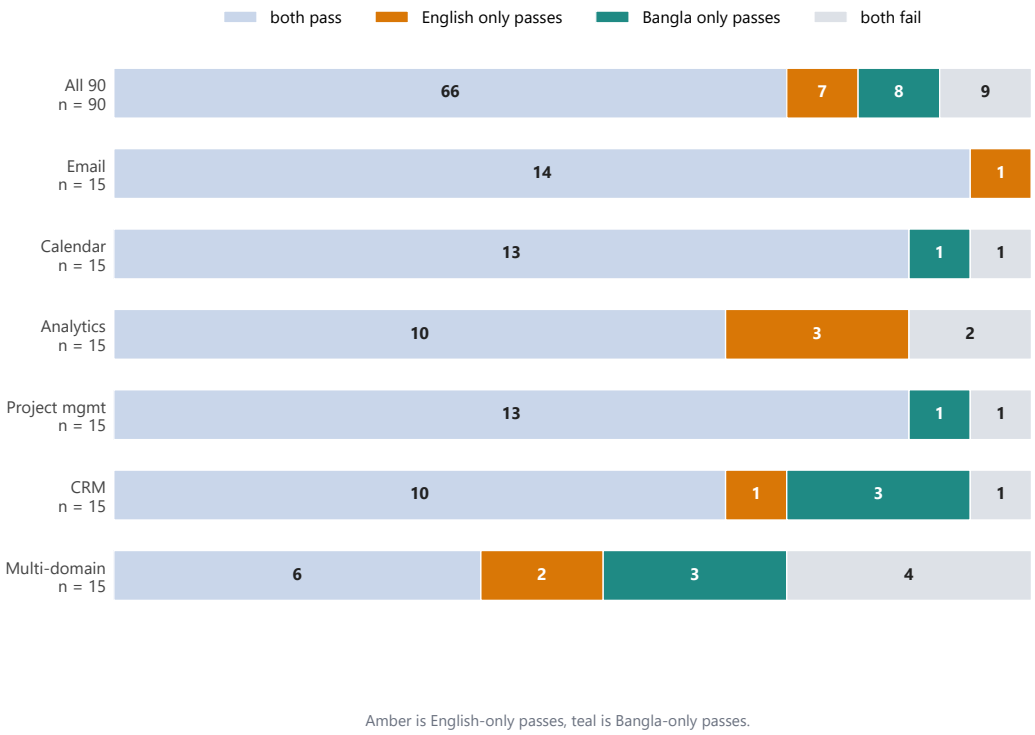


Figure 3. Success rate of the four runs by domain.

- gemma4:31b passes **81.1%** in English and **82.2%** in Bangla; gpt-oss:20b passes **58.9%** and **53.3%**.
- The two models differ much more than the two languages: gemma is ahead of gpt-oss by 22.2 pp in English and 28.9 pp in Bangla.
- The difference between the two models' language gaps is -6.7 pp (95% interval [-20.0, 6.7] pp): unlike OfficeBench, the models do not move in clearly opposite directions here.
- The largest one-way domain gap is gemma analytics (+20.0 pp, 3 against 0 flips); with 15 tasks per domain it is descriptive only.

R.3 gemma4:31b, English against Bangla, task by task

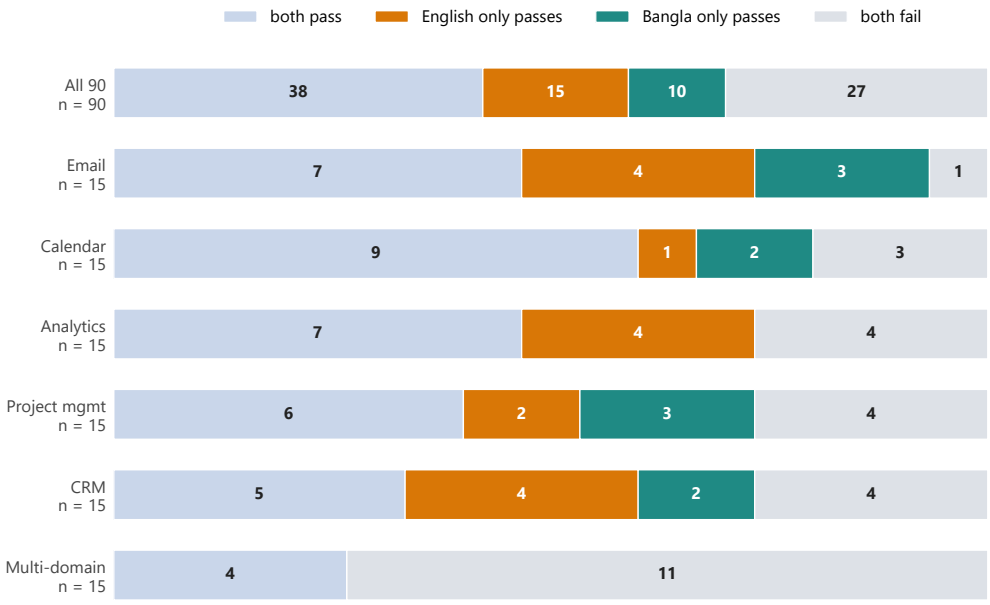


Amber is English-only passes, teal is Bangla-only passes.

Figure 4. Paired outcome of gemma4:31b English and Bangla on the same tasks.

- 7 tasks pass only in English and 8 only in Bangla; 66 pass in both and 9 fail in both.
- The English-only group is the one analysed case by case in Sections 4–12; the Bangla-only group is labelled on its English side, so confounds can be removed on both sides (Section 11).

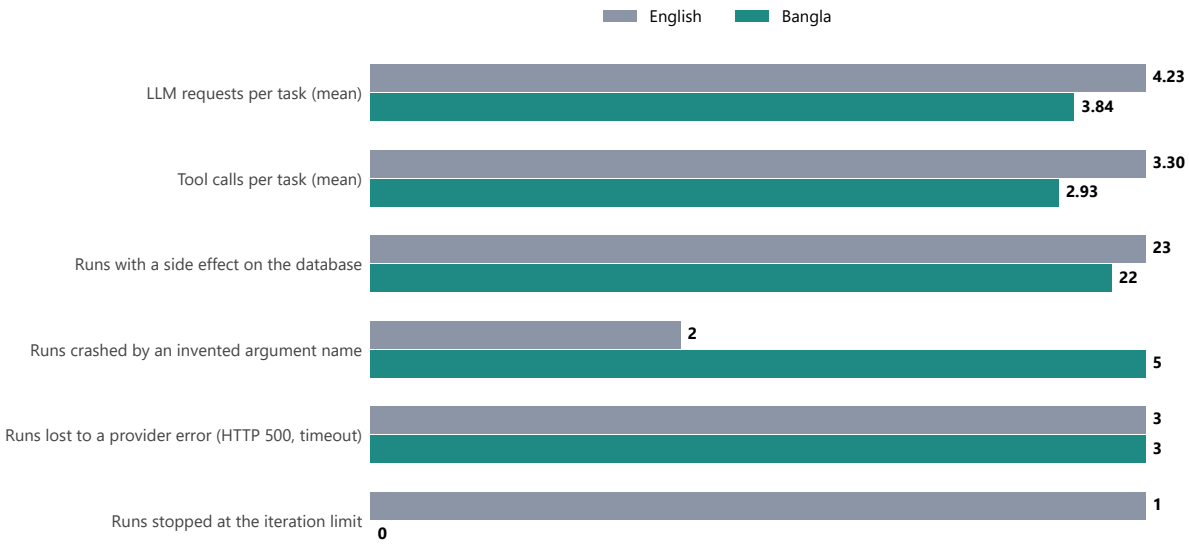
R.4 gpt-oss:20b, English against Bangla, task by task



Amber is English-only passes, teal is Bangla-only passes.

Figure 5. Paired outcome of gpt-oss:20b English and Bangla on the same tasks.

- 15 tasks pass only in English and 10 only in Bangla; 38 pass in both and 27 fail in both.
- The English-only group is the one analysed case by case in Sections 4–12; the Bangla-only group is labelled on its English side, so confounds can be removed on both sides (Section 11).

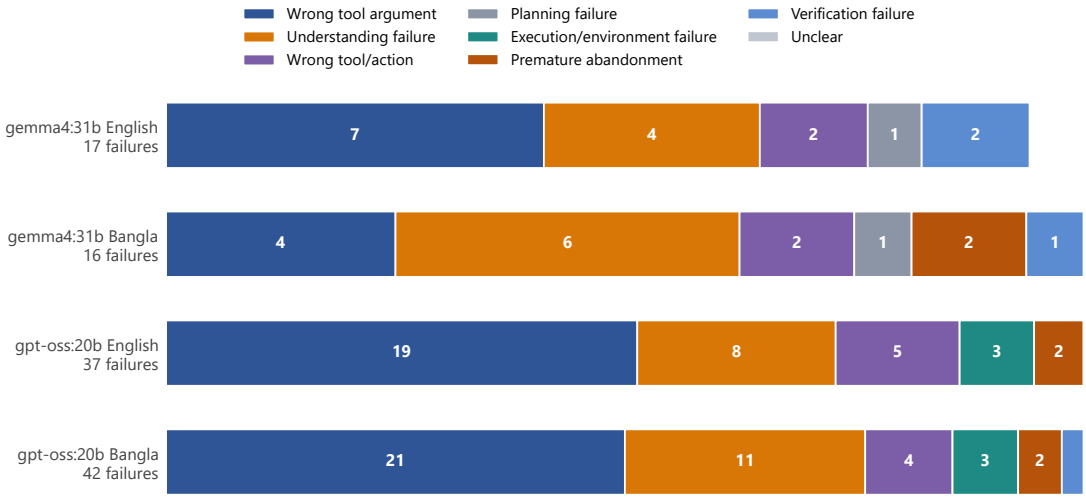


90 runs per column. Each row is scaled to its own larger value.

Figure 6. gpt-oss:20b English against Bangla: effort, side effects and how runs crash.

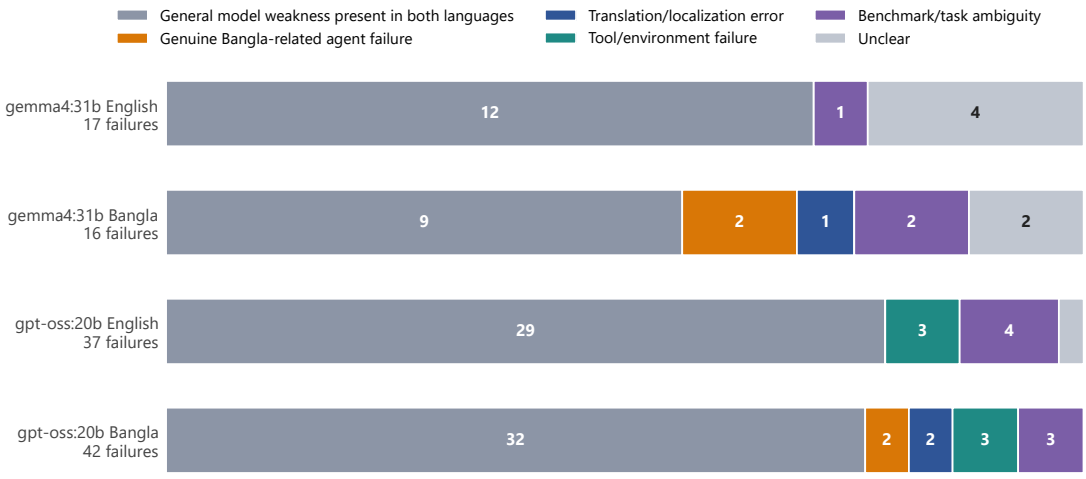
For comparison, gemma4:31b uses 3.78 requests and 3.96 tool calls per task in English against 3.29 and 3.17 in Bangla: its Bangla runs are leaner, largely because two English analytics runs looped on the same read call (analytics:76 and analytics:98; details in `docs/workbench/WORKBENCH_LOG_ANALYSIS.md`).

R.5 Why runs fail



Counts inside segments; widths are shares of each run's failures (failure type at the first failure point, adjudicated labels).

Figure 7. Failure type of every failed run, for all four runs (every failure is labelled).



Confound class of every failed run. On English failures the class says whether the same failure is language-independent.

Figure 8. Confound class of every failed run.

- **gpt-oss fails on tool arguments**: 40 of its 79 labelled failures are wrong tool arguments, in both languages.
- **gemma fails less and more evenly**: 33 labelled failures across both runs, led by wrong tool argument (11).
- **General weakness dominates every run**: 82 of 112 failed runs are coded C6 (the same mechanism appears in the other language).

R.6 What fails everywhere

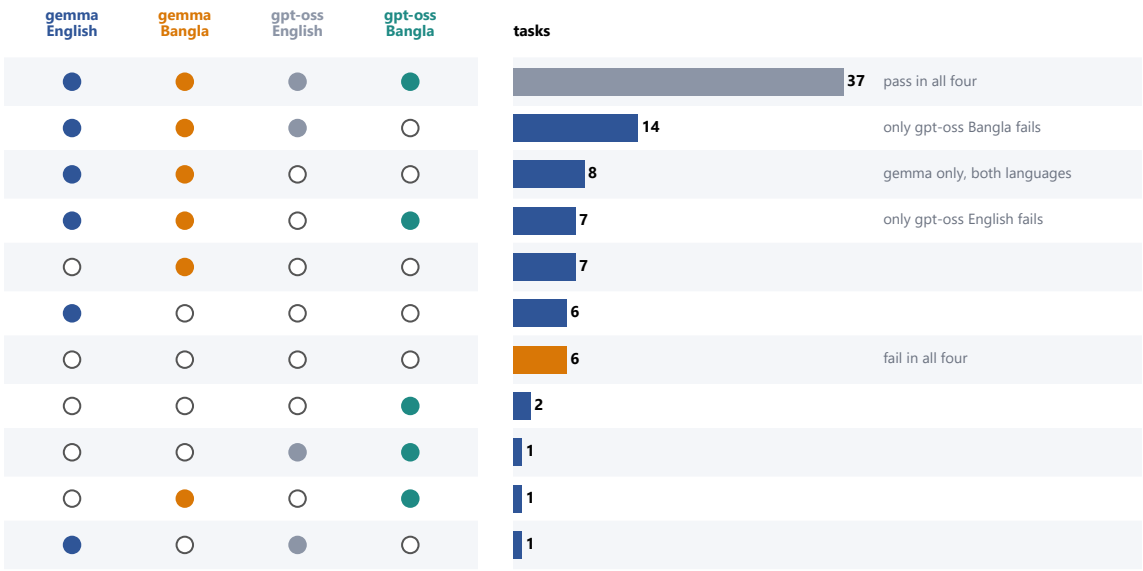


Figure 9. How the four runs agree on each task. Filled dot = passes, open dot = fails; rows sorted by number of tasks.

- **6** tasks fail in all four runs and **37** pass in all four.
- **8** tasks pass for gemma in both languages and fail for gpt-oss in both: the model, not the language, decides them.

---

1. Data prepared (pipeline section 1)

Every task has one row per model in `task_table.csv`. Alignment check: **90 of 90 task IDs appear in both runs of both models**; pass/fail in the task table equals `correct` in the per-task run tables for all 360 rows (asserted by this script).

Field	Where it comes from
dataset, model, task ID, run ID	WorkBench / model key / `workbench:<domain>:<n>` / run IDs in `run_meta_ref.json`, `run_meta_treat.json`
English and Bangla pass/fail	`results/comparisons/pilot_<model>_c6_vs_c0/paired.csv` (`correct_ref`, `correct_treat`)
Trajectory paths	`WorkBench/data/results/{c0,c6}/pilot_{en,bn}/<model>/*_traces.json#index`
Final state	`task_table.csv`: `en_final_state`, `bn_final_state`, `gt_state` (state-changing calls)
Tool calls, requests, errors	`per_task_ref.csv`, `per_task_treat.csv`
Labels	`results/failure_pipeline/pilot_<model>_c6_vs_c0/labels_final.csv`; notes in `trace_notes.md`; rules in `annotation_rules.md`

2. Basic outcome (pipeline section 2)

Model	Setting	n	EN pass	BN pass	Gap (EN−BN)	MELR	PP	PF	FP	FF
gemma4:31b	Overall	90	73 / 90 (81.1%)	74 / 90 (82.2%)	-1.1 pp	-1.4%	66	**7**	8	9
	Email	15	15 / 15 (100.0%)	14 / 15 (93.3%)	+6.7 pp	6.7%	14	**1**	0	0
	Calendar	15	13 / 15 (86.7%)	14 / 15 (93.3%)	-6.7 pp	-7.7%	13	**0**	1	1
	Analytics	15	13 / 15 (86.7%)	10 / 15 (66.7%)	+20.0 pp	23.1%	10	**3**	0	2
	Project mgmt	15	13 / 15 (86.7%)	14 / 15 (93.3%)	-6.7 pp	-7.7%	13	**0**	1	1
	CRM	15	11 / 15 (73.3%)	13 / 15 (86.7%)	-13.3 pp	-18.2%	10	**1**	3	1
	Multi-domain	15	8 / 15 (53.3%)	9 / 15 (60.0%)	-6.7 pp	-12.5%	6	**2**	3	4
gpt-oss:20b	Overall	90	53 / 90 (58.9%)	48 / 90 (53.3%)	+5.6 pp	9.4%	38	**15**	10	27
	Email	15	11 / 15 (73.3%)	10 / 15 (66.7%)	+6.7 pp	9.1%	7	**4**	3	1
	Calendar	15	10 / 15 (66.7%)	11 / 15 (73.3%)	-6.7 pp	-10.0%	9	**1**	2	3
	Analytics	15	11 / 15 (73.3%)	7 / 15 (46.7%)	+26.7 pp	36.4%	7	**4**	0	4
	Project mgmt	15	8 / 15 (53.3%)	9 / 15 (60.0%)	-6.7 pp	-12.5%	6	**2**	3	4
	CRM	15	9 / 15 (60.0%)	7 / 15 (46.7%)	+13.3 pp	22.2%	5	**4**	2	4
	Multi-domain	15	4 / 15 (26.7%)	4 / 15 (26.7%)	+0.0 pp	0.0%	4	**0**	0	11

MELR = (EN − BN) / EN. PP = EN-pass/BN-pass, **PF = EN-pass/BN-fail**, FP = EN-fail/BN-pass, FF = EN-fail/BN-fail. Denominator is the number of aligned tasks in the row.

3. Which cases were analysed (pipeline section 3)

Model	Group	Count	How analysed here
gemma4:31b	EN-pass / BN-fail	7	**All 7** read trace-against-trace, full §4 note, blind second review and adjudication (Appendix A/B).
	EN-fail / BN-pass	8	All 8: short labels on the English failure, so confounds can be removed on both sides.
	EN-fail / BN-fail	9	All 9, both sides labelled (18 rows): the general-difficulty baseline.
	EN-pass / BN-pass	66	Counted only (stable success).
gpt-oss:20b	EN-pass / BN-fail	15	**All 15** read trace-against-trace, full §4 note, blind second review and adjudication (Appendix A/B).
	EN-fail / BN-pass	10	All 10: short labels on the English failure, so confounds can be removed on both sides.
	EN-fail / BN-fail	27	All 27, both sides labelled (54 rows): the general-difficulty baseline.
	EN-pass / BN-pass	38	Counted only (stable success).

4. Trace notes (pipeline section 4)

The failure analysis follows one procedure for every failed run (Figure 10 shows it on a real task; Figure 11 is the coding scheme). One note per EN-pass/BN-fail case, with the nine fields the pipeline asks for; all notes are in Appendices A and B. Two worked examples show the level of detail.

email:41 · C1 · Email

- **Task goal:** forward every email that anaya sent last week with subject 'Update on Board of Directors Conclave' to nadia. Ground truth forwards 00000346 and 00000120, sent 11-20 and 11-21. The ground-truth logic uses the calendar week 2023-11-20..26.
- **English:** looked up both addresses, then `search_emails(query=subject, date_min=2023-11-19, date_max=2023-11-26)` returned both emails. It forwarded both. Pass.
- **Bangla:** "গত সপ্তাহে anaya '...' নিয়ে যে ইমেইলগুলো পাঠিয়েছিল, সবগুলো nadia-কে ফরোয়ার্ড করে দিতে পারবে". It looked up both addresses correctly, then searched `date_min=2023-11-23, date_max=2023-11-29` and got [] twice (steps 2 and 3, subject queries); a third search in that window by anaya's address (step 4) returned only other emails (00000166, 00000456). An undated search (step 5) returned both target emails, but the agent rejected them as outside last week. Its final answer: "গত সপ্তাহে (২৩ নভেম্বর থেকে ২৯ নভেম্বর) ... কোনো ইমেইল খুঁজে পাওয়া যায়নি ... আপনি কি অন্য কোনো তারিখ ... চেক করতে চান?"
- **First divergence:** step 2. "গত সপ্তাহে" was grounded as the rolling 7 days 11-23..11-29, where English used the calendar week.
- **Final failed / missing action:** the two `forward_email` calls were never made.
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Root cause:** a Bangla relative-date phrase was mis-grounded to the wrong date window. The wrong window then justified abandoning the task even after the correct emails had been found.
- **Confound:** Genuine Bangla-related agent failure (C1)
- **Evidence:** In email:41, English passed by calling `forward_email` for 00000120 and 00000346 after searching 11-19..11-26. In Bangla, the agent grounded "গত সপ্তাহে" as 11-23..11-29, found nothing, and refused the two emails its own undated search returned, so `forward_email` was never called. This is labelled an entity/slot-grounding failure of a Bangla date phrase: the translation is faithful and no C0 analogue exists.
- **Adjudication:** the failure type is now **Understanding failure**, because the type follows the first failure point (convention C1). The first failure point (Entity or slot grounding), the confound (Genuine, provisional; decided together with md:120 under C5) and the tree are unchanged. Factual fix: the Bangla bullet originally said the windowed searches returned [] "three times"; the trace shows twice (steps 2 and 3), corrected above.

crm:77 · C6 · CRM

- **Task goal:** move customers who haven't responded to a software proposal in 6 weeks to lost (GT: no change; there are none).
- **English:** search_customers(product_interest="Software", status="Proposal", last_contact_date_max="2023-10-19") → [], so it changed nothing. Pass.
- **Bangla:** search_customers(product_interest="Software", last_contact_date_max="2023-10-19"), with no status filter, then set 5 customers with status Won/Qualified/Lead to Lost. Fail.
- **First divergence:** step 0, the status filter is missing.
- **Final failed / missing action:** five spurious update_customer(..., status, "Lost") calls.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** the "proposal" status constraint was dropped.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** `search_customers {'product_interest': 'Software', 'last_contact_date_max': '2023-10-19'}`, followed by 5 updates to Lost.
- **Adjudication:** failure type Understanding failure; confound General model weakness; tree S1 ok · S3 fail · S4 fail · S5 na · S7 ok. The transliteration keeps the word's meaning, and the English 'a proposal' admits the same drop (C9, C5(1)). The same model mapped the same 'প্রপোজালে' correctly in crm:74 BN. A status filter missing from the first search also fails C0 crm:42 EN and md:182 EN. The policy deviation remains flagged for the native-speaker review.

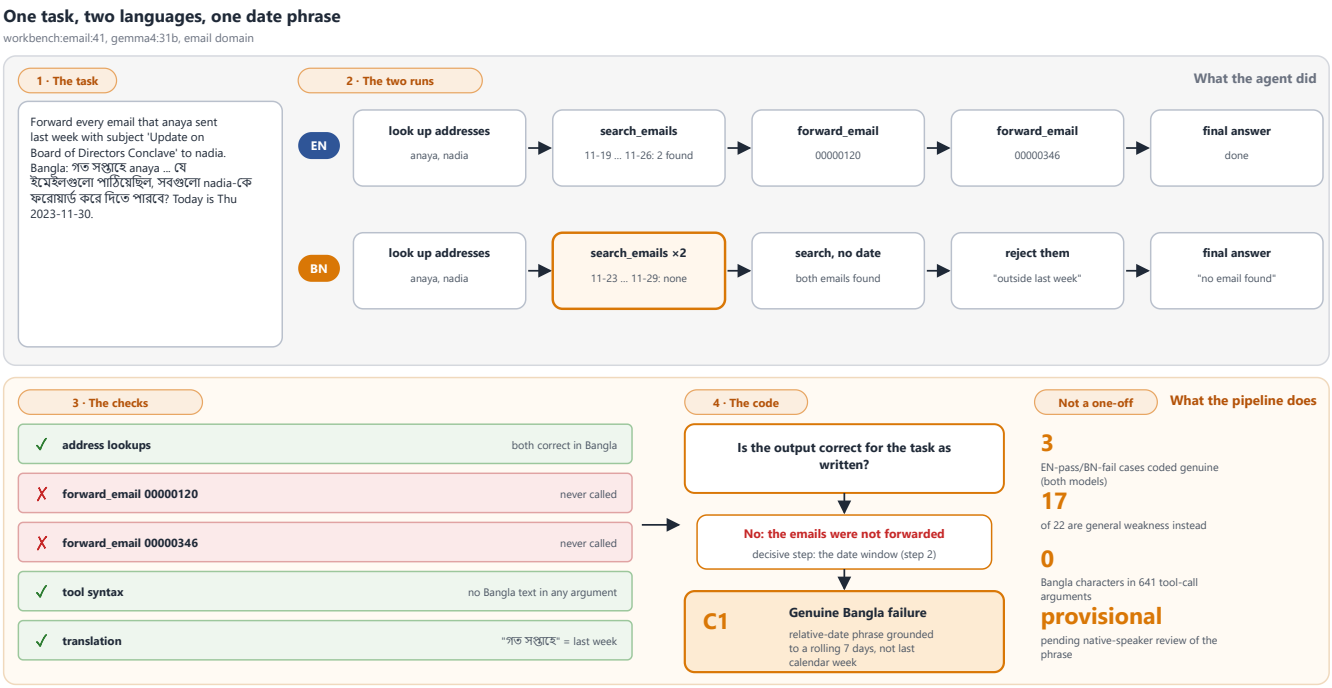


Figure 10. One real task carried through the pipeline: the Bangla run grounds "last week" to a different window, then rejects the right emails.

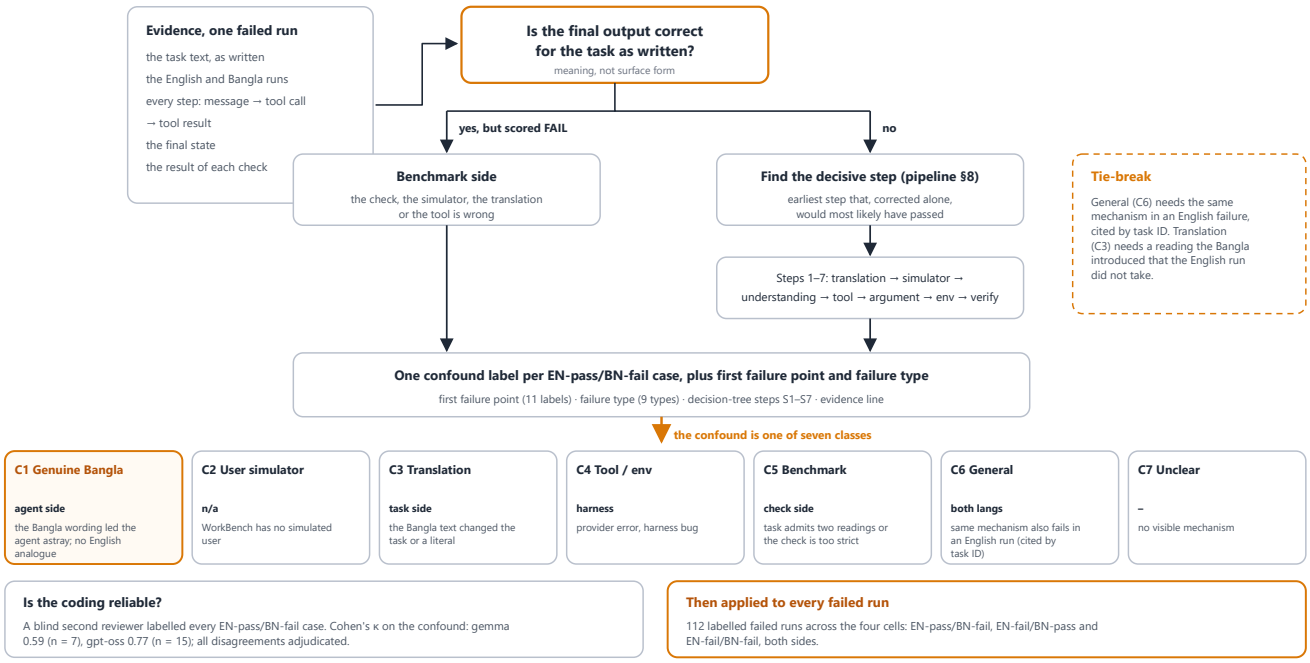


Figure 11. How failures are characterized: one question, one decisive step, one confound class per case.

5. First failure point (pipeline section 5)

Rule applied: label the earliest upstream point after which the correct final state became unreachable, not the visible final error. Denominator: EN-pass/BN-fail cases of each model.

First failure point	gemma	share of 7	gpt-oss	share of 15
Instruction understanding	1	14.3%	0	0.0%
Entity or slot grounding	2	28.6%	1	6.7%
Task constraint preservation	1	14.3%	1	6.7%
Planning	1	14.3%	0	0.0%
Tool selection	0	0.0%	1	6.7%
Tool-argument construction	2	28.6%	11	73.3%
Premature termination	0	0.0%	1	6.7%

Reading the table: gpt-oss's Bangla-only failures start mostly at **tool-argument construction** (11 of 15); gemma's spread over grounding, constraint preservation and instruction understanding. No case starts at tool selection for gemma.

6. Failure type (pipeline section 6)

Failure type	gemma	share of 7	gpt-oss	share of 15
Understanding failure	3	42.9%	1	6.7%
Planning failure	1	14.3%	0	0.0%
Wrong tool/action	1	14.3%	2	13.3%
Wrong tool argument	2	28.6%	11	73.3%
Premature abandonment	0	0.0%	1	6.7%

Under adjudication convention C1 the failure type names the failure *at* the first failure point, so the two distributions are one judgment, not two independent ones. The pipeline's watch items map as follows: premature STOP and dialogue-state failure have no equivalent (single-turn); internal ID used as a name does not occur; changed task constraint is the date-threshold and status-filter cases (task constraint preservation); compound task partially completed appears in multi-domain tasks.

7. Confound type (pipeline section 7)

Confound	gemma	share of 7	gpt-oss	share of 15
C1 Genuine Bangla-related agent failure	2	28.6%	1	6.7%
C3 Translation/localization error	1	14.3%	0	0.0%
C4 Tool/environment failure	0	0.0%	0	0.0%
C5 Benchmark/task ambiguity	1	14.3%	0	0.0%
C6 General model weakness present in both languages	3	42.9%	14	93.3%
C7 Unclear	0	0.0%	0	0.0%

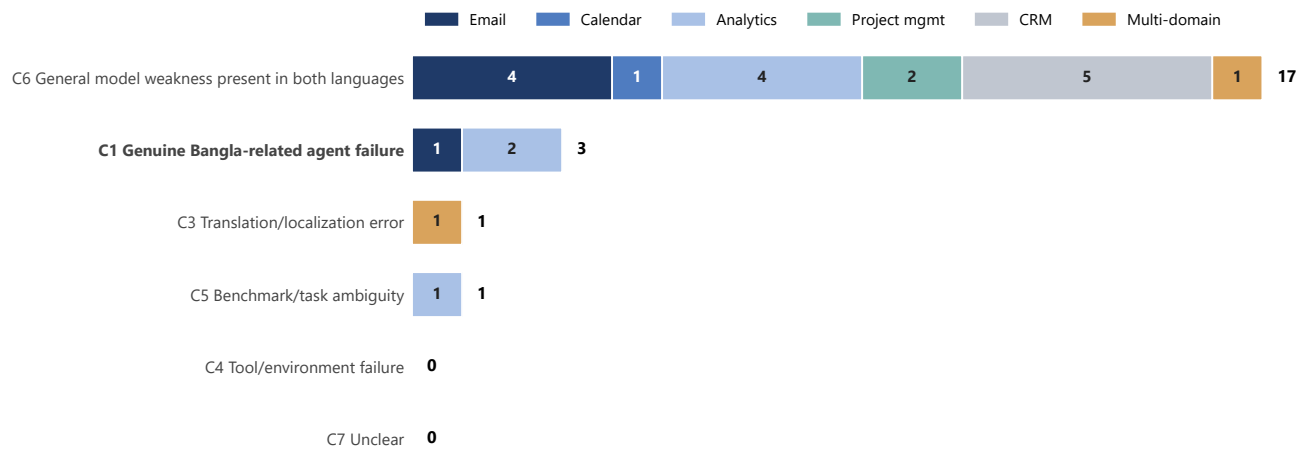


Figure 12. Confound class of the EN-pass/BN-fail cases of both models, split by domain.

C2 (user simulator) is zero by construction. Both models together: genuine 3, translation 1, tool/environment 0, benchmark ambiguity 1, general 17, unclear 0.

8. Root-cause decision tree (pipeline section 8)

Each EN-pass/BN-fail case leaves the tree at the first step that fails (field `decision_tree`). Counts of exits:

Exit step	gemma	gpt-oss
Step 1: translated task not valid	1	0
Step 2: user simulator (not applicable)	0	0
Step 3: agent misunderstood / mis-grounded	2	1
Step 4: wrong tool/action	2	2
Step 5: wrong argument	2	11
Step 6: tool/environment	0	0
Step 7: verification / abandonment	0	1

Step 2 never fires (no user simulator). The pipeline's added Step 0 of the OfficeBench report (output already correct, the check rejects it) is covered here by C5 benchmark ambiguity.

9. Evidence for each root cause (pipeline section 9)

Each labelled row carries the English outcome, the Bangla outcome, the first failure point, the tool call or message that shows it, why it is or is not a general model failure (`not_general_reason`), and the confound label. Full notes are in Appendices A and B.

10. Counts (pipeline section 10)

Count	gemma	gpt-oss	both	share of 22
Genuine Bangla-related agent failures (C1), after filtering	2	1	3	13.6%
Bangla user-simulator-caused (C2)	0	0	0	0.0%
Translation-caused (C3)	1	0	1	4.5%
Tool/environment-caused (C4)	0	0	0	0.0%
Benchmark/task ambiguity (C5)	1	0	1	4.5%
General weakness present in both (C6)	3	14	17	77.3%
Unclear, candidates for genuine failure (C7)	0	0	0	0.0%

Model-wise: gemma 7 cases, gpt-oss 15. Domain-wise: Figure 12. The upper bound on genuine Bangla effects is C1 + C7 = 3 cases (13.6% of 22); the lower bound is C1 = 3 (13.6%).

11. Statistical analysis (pipeline section 11)

Paired design: the same 90 tasks in both languages. McNemar exact two-sided test on discordant pairs; 95% intervals: Wilson for proportions, paired bootstrap (4000 resamples of tasks, seed 7) for the gap and MELR. Effect size: Cohen's g = b/(b+c) − 0.5 and the discordant odds ratio b/c. Holm across the two models.

Model	EN [95% CI]	BN [95% CI]	Gap [95% CI]	MELR [95% CI]	b / c	McNemar p	Holm p	Cohen's g	Odds b/c
gemma4:31b	81.1% [71.8, 87.9]	82.2% [73.1, 88.8]	-1.1 pp [-10.0, 7.8]	-1.4% [-12.7, 8.9]	7 / 8	1.000	1.000	-0.03	0.88
gpt-oss:20b	58.9% [48.6, 68.5]	53.3% [43.1, 63.3]	+5.6 pp [-5.6, 16.7]	9.4% [-10.6, 25.8]	15 / 10	0.424	0.849	0.10	1.50

Neither model has a detectable gap after or before Holm. gemma's gap is -1.1 pp with interval [-10.0, 7.8] pp; gpt-oss's is +5.6 pp with interval [-5.6, 16.7] pp, so a Bangla cost of up to 8 (gemma) or 17 (gpt-oss) points cannot be excluded either: the pilot is too small to show equivalence.

Sensitivity analysis

Unlike the OfficeBench analysis, the EN-fail/BN-pass side is labelled with the same rules, so every scenario removes confounded tasks **symmetrically**: a task is dropped if its Bangla failure (in an EN-pass/BN-fail pair) or its English failure (in an EN-fail/BN-pass pair) carries the confound.

gemma4:31b

Scenario	n	EN	BN	Gap	Gap 95% CI (pp)	MELR	PF / FP	McNemar p
S0 All 90 tasks (as scored)	90	81.1%	82.2%	-1.1 pp	[-10.0, 7.8]	-1.4%	7 / 8	1.000
S1 Drop translation-confounded tasks (C3)	89	80.9%	83.1%	-2.2 pp	[-11.2, 5.6]	-2.8%	6 / 8	0.791
S2 Drop tool/environment-confounded tasks (C4)	90	81.1%	82.2%	-1.1 pp	[-10.0, 7.8]	-1.4%	7 / 8	1.000
S3 Drop benchmark-ambiguity tasks (C5)	89	80.9%	83.1%	-2.2 pp	[-10.1, 5.6]	-2.8%	6 / 8	0.791
S4 Drop C3 + C4 + C5	88	80.7%	84.1%	-3.4 pp	[-11.4, 4.5]	-4.2%	5 / 8	0.581
S5 Drop C3 + C4 + C5 + C6 (leave C1 and C7)	80	85.0%	86.2%	-1.3 pp	[-6.2, 3.8]	-1.5%	2 / 3	1.000
S6 Drop tasks with a provider error on either side	90	81.1%	82.2%	-1.1 pp	[-10.0, 7.8]	-1.4%	7 / 8	1.000

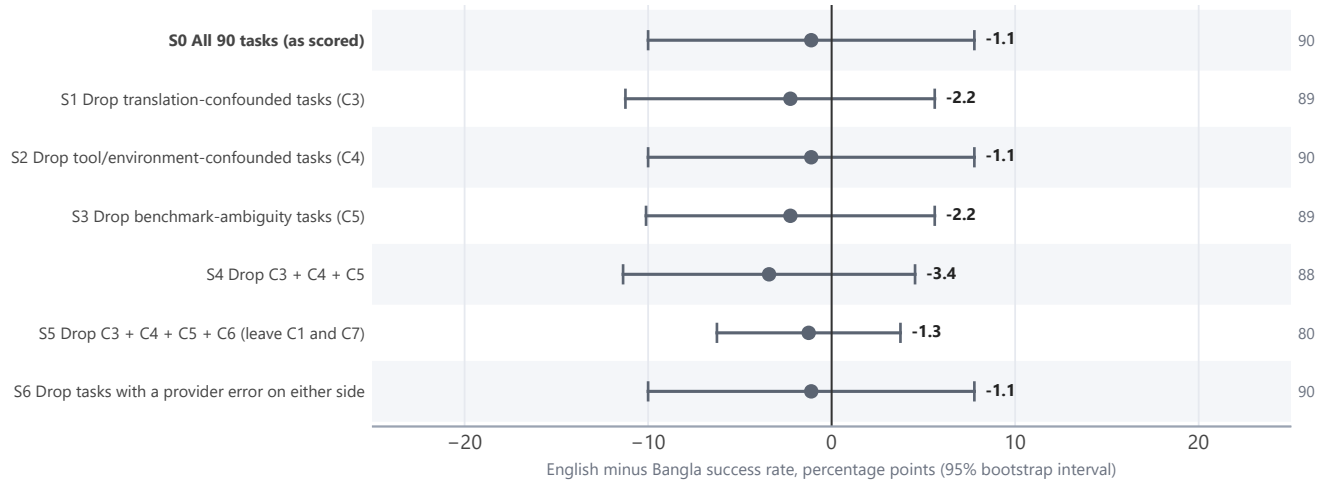


Figure 13. gemma4:31b: English minus Bangla gap with 95% interval under each sensitivity scenario. Amber intervals exclude zero.

gpt-oss:20b

Scenario	n	EN	BN	Gap	Gap 95% CI (pp)	MELR	PF / FP	McNemar p
S0 All 90 tasks (as scored)	90	58.9%	53.3%	+5.6 pp	[-5.6, 16.7]	9.4%	15 / 10	0.424
S1 Drop translation-confounded tasks (C3)	90	58.9%	53.3%	+5.6 pp	[-5.6, 16.7]	9.4%	15 / 10	0.424
S2 Drop tool/environment-confounded tasks (C4)	89	59.6%	52.8%	+6.7 pp	[-3.4, 16.9]	11.3%	15 / 9	0.307
S3 Drop benchmark-ambiguity tasks (C5)	89	59.6%	52.8%	+6.7 pp	[-3.4, 18.0]	11.3%	15 / 9	0.307
S4 Drop C3 + C4 + C5	88	60.2%	52.3%	+8.0 pp	[-2.3, 18.2]	13.2%	15 / 8	0.210
S5 Drop C3 + C4 + C5 + C6 (leave C1 and C7)	66	59.1%	57.6%	+1.5 pp	[0.0, 4.5]	2.6%	1 / 0	1.000
S6 Drop tasks with a provider error on either side	85	62.4%	55.3%	+7.1 pp	[-3.6, 17.6]	11.3%	15 / 9	0.307

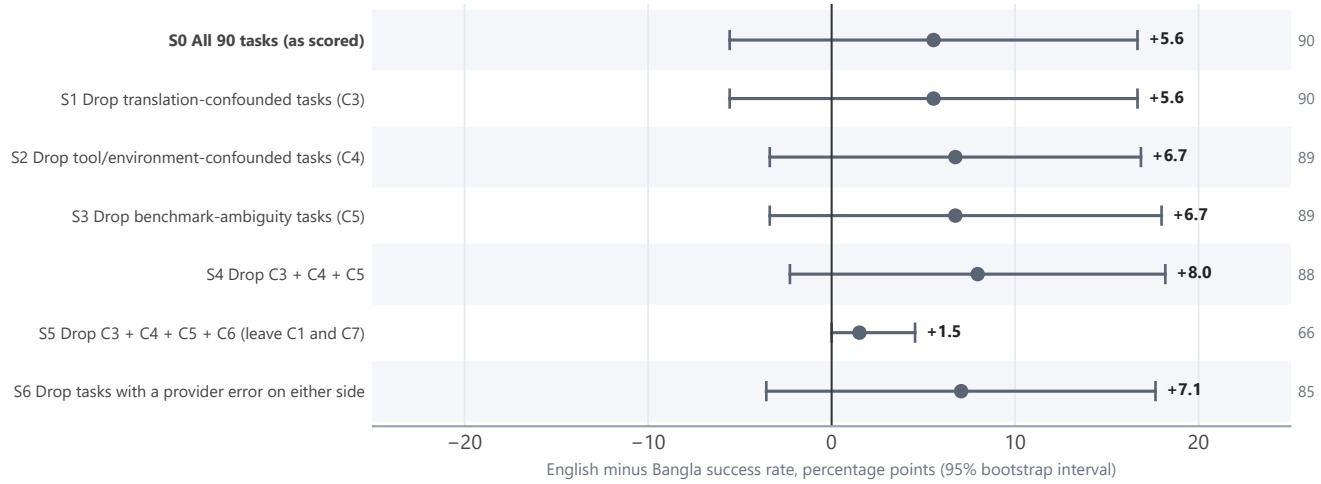


Figure 14. gpt-oss:20b: English minus Bangla gap with 95% interval under each sensitivity scenario. Amber intervals exclude zero.

How to read this. No scenario moves either model to a significant gap: the confound classes are small, and removing them on both sides changes the gap by a few points at most. S5 keeps only the genuine and unclear cases on the Bangla side; it is the closest to a "pure Bangla agent effect" and its interval still includes zero.

12. Findings (pipeline section 12)

Each finding gives observation, evidence, root cause and boundary.

F1. WorkBench shows no detectable English–Bangla gap for either model. gemma 81.1% vs 82.2% (p = 1.000); gpt-oss 58.9% vs 53.3% (p = 0.424); the flips go both ways (7/8 and 15/10). *Boundary:* 90 tasks and one run per cell; the intervals allow a Bangla cost of up to about 8–17 points.

F2. Most Bangla-only failures are general model weaknesses. 17 of 22 EN-pass/BN-fail cases have the same mechanism in an English failure of the same model, cited by task ID (date thresholds, missing status filters, wrong free slot). *Boundary:* the C6 rule needs only one English analogue, so it can over-assign general weakness.

F3. The genuine Bangla mechanism is relative-date grounding. 3 cases are coded C1. The clearest (email:41, gemma) grounds "গত সপ্তাহে" (last week) to the rolling 7 days instead of the last calendar week, then rejects the right emails its own search returned. Tool syntax stays correct and no Bangla character enters a tool argument. *Boundary:* provisional until the native-speaker review of the date phrases; if the phrase leans to "the past 7 days", the label becomes translation (C3).

F4. gpt-oss's failures are tool-argument failures in both languages. 40 of its 79 labelled failed runs are wrong tool arguments; invented argument names (`new?`, `field`, `value_to_plot`) crash runs in both languages. *Boundary:* 5 such crashes in Bangla against 2 in English is not significant.

F5. Provider errors do not drive the gpt-oss gap. Dropping every task with an HTTP 500 or timeout on either side (S6) leaves the gap almost unchanged and non-significant. *Boundary:* the gpt-oss runs were resumed over two days against a shared provider.

Not claimed. The data do not support "Bangla agents reason worse" on WorkBench: the Bangla runs are as accurate as the English ones for both models, and gemma's Bangla runs are leaner.

13. Paper outputs (pipeline section 13)

#	Output	Status	Where
1	Overall English vs Bangla success table	Done	Section R, Section 2
2	MELR by dataset and model	Done (one dataset, two models, by domain)	Section 2, Section 11
3	Paired outcome table	Done	Section 2 (PP / PF / FP / FF), Figures 4–5
4	Failure-type distribution for PF	Done	Section 6
5	First-failure-point distribution	Done	Section 5
6	Confound filtering table	Done	Section 7, Section 10
7	Before/after confound-filtered results	Done (symmetric)	Section 11, sensitivity
8	Two or three short case studies	Done	Section 15
9	Appendix with annotation rules and long trajectories	Notes done; rules in `annotation_rules.md`	Appendices A–C

14. Quality-control checklist (pipeline section 14)

Check	Status	Note
Every task ID aligned between English and Bangla	Pass	90 of 90 for both models; pass/fail cross-checked against the per-task tables.
Every metric has a denominator	Pass	Stated in each table header or caption.
EN-pass/BN-fail separated from both-fail	Pass	Sections 2, 3, R.5.
Translation errors not counted as genuine Bangla failures	Pass	C3 is its own class.
User-simulator errors counted separately	Not applicable	No user simulator in WorkBench.
Tool/environment failures counted separately	Pass	C4 is its own class; provider errors also handled in S6.
A second reviewer checks a subset	Pass	Blind A2 on every EN-pass/BN-fail case. κ (first failure point / type / confound): gemma 0.45 / 0.48 / 0.59; gpt-oss 0.91 / 0.76 / 0.77.
Disagreements adjudicated	Pass	Adjudicator ADJ; reasons in `adjudication.csv`; conventions C1–C12 in `annotation_rules.md`.
Case studies match the quantitative patterns	Pass	Section 15 cases come from the C1, C6 and C5 classes.
Strong claims only after confound filtering	Pass	F1–F5 each state a boundary; symmetric sensitivity in Section 11.

Other limits: one run per cell at temperature 0, so run-to-run variation is not measured; English and Bangla ran on different days against a shared provider; all annotators are Claude models; κ is measured before adjudication and is not the reliability of the final labels.

15. Case studies

email:41 (gemma4:31b): A date phrase grounded to the wrong week. *English:* looked up both addresses, then `search_emails(query=subject, date_min=2023-11-19, date_max=2023-11-26)` returned both emails. It forwarded both. Pass. *Bangla:* "গত সপ্তাহে anaya '...' নিয়ে যে ইমেইলগুলো পাঠিয়েছিল, সবগুলো nadia-কে ফরোয়ার্ড করে দিতে পারবে". It looked up both addresses correctly, then searched `date_min=2023-11-23, date_max=2023-11-29` and got [] twice (steps 2 and 3, subject queries); a third search in that window by anaya's address (step 4) returned only other emails (00000166, 00000456). An undated search (step 5) returned both target emails, but the agent rejected them as outside last week. Its final answer: "গত সপ্তাহে (২৩ নভেম্বর থেকে ২৯ নভেম্বর) ... কোনো ইমেইল খুঁজে পাওয়া যায়নি ... আপনি কি অন্য কোনো তারিখ ... চেক করতে চান?"

Root cause: a Bangla relative-date phrase was mis-grounded to the wrong date window. The wrong window then justified abandoning the task even after the correct emails had been found. *Label:* C1, Genuine Bangla-related agent failure, first failure point "Entity or slot grounding".

crm:77 (gemma4:31b): The same wrong decision, through a tool that does not exist. *English:* `search_customers(Software, Proposal)` returned 5 customers. It set 00000147 (last contact 2023-10-29) to Lost, then set it back to Proposal twice, and answered "no customers have been inactive for 6 weeks or more (cutoff date: 2023-10-19)". The net state is unchanged. Pass. *Bangla:* the same search. At step 1 it set 00000147 to Lost through a **nonexistent tool**, `customer_relationship_manager_update_task(task_id=00000147, …)`. The DB was unchanged, but the call stays in `function_calls`, `_parse_and_dispatch` rejects it, and `is_correct` requires every call to execute. **That alone makes the run fail.** It then called `update_customer(00000147, status, Lost)`, with the claim "৬ সপ্তাহ বা তার বেশি সময় ধরে কোনো যোগাযোগ হয়নি ... (শেষ যোগাযোগ ছিল ২০২৩-১০-২৯)". There was no revert.

Root cause: the 6-week threshold was misapplied (32 days treated as ≥ 42). This decision is upstream of both bad calls. A secondary, independently fatal error is the invalid tool name. It is the only unrecognised call in either run (0/90 C0), and it shows no visible Bangla link, because tool descriptions are English in C6. *Label:* C6, General model weakness present in both English and Bangla, first failure point "Task constraint preservation".

analytics:98 (gemma4:31b): An English pass that only looped its way to the answer. *English:* 24 read calls cycling over 11-29/11-30, then "I was unable to calculate the percent growth because there is no data available for today". No plot was made. This passed by abstention, not by computing the answer. *Bangla:* queried 11-29 (20.5 s, 10 visits) and then **11-28** (20.32 s, 19 visits). It computed session +0.9% against visits −47%, judged the condition true, and called `create_plot(line, time_min=2023-11-28)` for both metrics. Its answer correctly says "বুধবার (২০২৩-১১-২৯) থেকে".

Root cause: the task is degenerate. "Growth since yesterday", with data only through yesterday, has no meaningful value, and the ground truth encodes "0 vs 0 → no action". Bangla resolved the ill-posed task one plausible way; English gave up. *Label:* C5, Benchmark/task ambiguity, first failure point "Planning".

Appendix A. Trace notes for the 7 gemma4:31b EN-pass/BN-fail cases

Ordered by task ID. Fields follow pipeline section 4. Labels (first failure point, type, confound) are the adjudicated ones from `labels_final.csv`; narrative fields come from `trace_notes.md`.

analytics:119 · C6 · Analytics

- **Task goal:** if social media traffic exceeded search engine traffic over the last 4 weeks, make bar charts of both. Ground truth: `time_min=2023-11-02`.
- **English:** traffic counts from 11-02, then two bar `create_plot` calls with `time_min=2023-11-02`. Pass. EN's `time_max=11-30` is accepted by `accept_either_chart_end_date`.
- **Bangla:** "গত 4 সপ্তাহের social media আর search engine-এর bar চার্ট বানাও, ...". Traffic counts and two bar plots, both with `time_min=2023-10-26`, which is 5 weeks.
- **First divergence:** step 0, the time_min of the very first read call.
- **Final failed / missing action:** both `create_plot` calls with `time_min=2023-10-26`.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** the Bangla N-week window was mis-grounded, off by one week.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** In analytics:119, English passed by calling `create_plot(bar)` for both sources with `time_min=2023-11-02`. In Bangla, the agent grounded "গত 4 সপ্তাহের" to `time_min=2023-10-26`, so both plots cover 5 weeks. This is labelled a Bangla date/quantity grounding failure: the translation is faithful and no C0 analogue exists.
- **Adjudication:** first failure point is now **Tool-argument construction**, the confound is **General model weakness present in both English and Bangla**, and the tree has S3 ok, so the first fail is S5. The failure type (Wrong tool argument) is unchanged. Reason (C4):

analytics:20 · C1 · Analytics

- **Task goal:** plot the distribution of engaged users and total visits for Nov 5–21. Ground truth: two `create_plot(..., plot_type="histogram")` calls.
- **English:** two histogram `create_plot` calls with the correct dates. Pass.
- **Bangla:** "প্লিজ 5 নভেম্বর থেকে 21 নভেম্বর সময়ের মধ্যে এনগেজড ইউজার ও মোট ভিজিটের ডিস্ট্রিবিউশন আমাকে চার্টে দেখিয়ে দাও". It made two `create_plot(..., plot_type="line")` calls with the correct dates and metrics, and described them as "ডিস্ট্রিবিউশন চার্টগুলো".
- **First divergence:** step 0, the choice of plot_type.
- **Final failed / missing action:** `create_plot` with `line` instead of `histogram`, twice.
- **First failure point:** Instruction understanding · **Type:** Understanding failure
- **Root cause:** the Bangla request for a "distribution" chart over a date range was not mapped to a histogram.
- **Confound:** Genuine Bangla-related agent failure (C1)
- **Evidence:** In analytics:20, English passed by calling `create_plot` with `plot_type="histogram"` for both metrics. In Bangla, the agent called `create_plot` with `plot_type="line"` for the same request, phrased "ডিস্ট্রিবিউশন ... চার্টে দেখিয়ে দাও". This is labelled an instruction-understanding failure tied to the Bangla wording. It is not a general weakness (no C0 analogue), but the evidence is weak (analytics:15 BN succeeded).
- **Adjudication:** the failure type is now **Understanding failure**, because the type follows the first failure point (C1). A2's Tool-argument construction was rejected: 'line' is what a loose reading of "distribution over Nov 5–21" gives, and the agent calls its line charts distribution charts. That is a different meaning, not a mis-mapping of an understood value (C4). The first failure point (Instruction understanding), the confound (Genuine, weak) and the tree are unchanged.

analytics:98 · C5 · Analytics

- **Task goal:** "percent growth of average session duration since Wednesday; if it grew by more than total visits, plot both lines since then".
- **English:** 24 read calls cycling over 11-29/11-30, then "I was unable to calculate the percent growth because there is no data available for today". No plot was made. This passed by abstention, not by computing the answer.
- **Bangla:** queried 11-29 (20.5 s, 10 visits) and then **11-28** (20.32 s, 19 visits). It computed session +0.9% against visits −47%, judged the condition true, and called `create_plot(line, time_min=2023-11-28)` for both metrics. Its answer correctly says "বুধবার (২০২৩-১১-২৯) থেকে".
- **First divergence:** step 2. To get a "growth on Wednesday", the agent invented a Tuesday baseline.
- **Final failed / missing action:** two `create_plot` calls that should not exist.
- **First failure point:** Planning · **Type:** Planning failure
- **Root cause:** the task is degenerate. "Growth since yesterday", with data only through yesterday, has no meaningful value, and the ground truth encodes "0 vs 0 → no action". Bangla resolved the ill-posed task one plausible way; English gave up.
- **Confound:** Benchmark/task ambiguity (C5)
- **Evidence:** In analytics:98, English passed only by abstaining ("unable to calculate … no data available for today") after looping reads. In Bangla, the agent compared 11-28 with 11-29, declared the condition met, and called `create_plot(line)` twice. Ground truth needs no plot because growth since the last data day is 0 for both metrics. This is labelled a planning failure under benchmark/task ambiguity, not a Bangla effect.

crm:77 · C6 · CRM

- **Task goal:** move software-proposal customers who have not responded in 6 weeks to Lost. Ground truth: **no change**, because no candidate's last contact is before the 2023-10-19 cutoff.
- **English:** `search_customers(Software, Proposal)` returned 5 customers. It set 00000147 (last contact 2023-10-29) to Lost, then set it back to Proposal twice, and answered "no customers have been inactive for 6 weeks or more (cutoff date: 2023-10-19)". The net state is unchanged. Pass.
- **Bangla:** the same search. At step 1 it set 00000147 to Lost through a **nonexistent tool**, `customer_relationship_manager_update_task(task_id=00000147, …)`. The DB was unchanged, but the call stays in `function_calls`, `_parse_and_dispatch` rejects it, and `is_correct` requires every call to execute. **That alone makes the run fail.** It then called `update_customer(00000147, status, Lost)`, with the claim "৬ সপ্তাহ বা তার বেশি সময় ধরে কোনো যোগাযোগ হয়নি … (শেষ যোগাযোগ ছিল ২০২৩-১০-২৯)". There was no revert.
- **First divergence:** step 1. The wrong decision is the same as English's step 1, but Bangla expressed it through an invalid tool name. Bangla therefore could not recover by reverting, as English did; it also never tried to.
- **Final failed / missing action:** the rejected `customer_relationship_manager_update_task` call, plus the kept `update_customer(00000147 → Lost)`.
- **First failure point:** Task constraint preservation · **Type:** Wrong tool/action
- **Root cause:** the 6-week threshold was misapplied (32 days treated as ≥ 42). This decision is upstream of both bad calls. A secondary, independently fatal error is the invalid tool name. It is the only unrecognised call in either run (0/90 C0), and it shows no visible Bangla link, because tool descriptions are English in C6.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** In customer_relationship_manager:77, English passed because it reverted its mistaken `update_customer(00000147 → Lost)`. In Bangla, the agent made the same decision, first through the nonexistent `customer_relationship_manager_update_task` (scored as a failed call) and then through `update_customer`, which it kept, claiming a 2023-10-29 contact was over 6 weeks old. This is a constraint-preservation (date-threshold) error. It is a general model weakness, because C0 customer_relationship_manager:74 fails by the same mechanism.

email:41 · C1 · Email

- **Task goal:** forward every email that anaya sent last week with subject 'Update on Board of Directors Conclave' to nadia. Ground truth forwards 00000346 and 00000120, sent 11-20 and 11-21. The ground-truth logic uses the calendar week 2023-11-20..26.
- **English:** looked up both addresses, then `search_emails(query=subject, date_min=2023-11-19, date_max=2023-11-26)` returned both emails. It forwarded both. Pass.
- **Bangla:** "গত সপ্তাহে anaya '...' নিয়ে যে ইমেইলগুলো পাঠিয়েছিল, সবগুলো nadia-কে ফরোয়ার্ড করে দিতে পারবে". It looked up both addresses correctly, then searched `date_min=2023-11-23, date_max=2023-11-29` and got [] twice (steps 2 and 3, subject queries); a third search in that window by anaya's address (step 4) returned only other emails (00000166, 00000456). An undated search (step 5) returned both target emails, but the agent rejected them as outside last week. Its final answer: "গত সপ্তাহে (২৩ নভেম্বর থেকে ২৯ নভেম্বর) ... কোনো ইমেইল খুঁজে পাওয়া যায়নি ... আপনি কি অন্য কোনো তারিখ ... চেক করতে চান?"
- **First divergence:** step 2. "গত সপ্তাহে" was grounded as the rolling 7 days 11-23..11-29, where English used the calendar week.
- **Final failed / missing action:** the two `forward_email` calls were never made.
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Root cause:** a Bangla relative-date phrase was mis-grounded to the wrong date window. The wrong window then justified abandoning the task even after the correct emails had been found.
- **Confound:** Genuine Bangla-related agent failure (C1)
- **Evidence:** In email:41, English passed by calling `forward_email` for 00000120 and 00000346 after searching 11-19..11-26. In Bangla, the agent grounded "গত সপ্তাহে" as 11-23..11-29, found nothing, and refused the two emails its own undated search returned, so `forward_email` was never called. This is labelled an entity/slot-grounding failure of a Bangla date phrase: the translation is faithful and no C0 analogue exists.
- **Adjudication:** the failure type is now **Understanding failure**, because the type follows the first failure point (convention C1). The first failure point (Entity or slot grounding), the confound (Genuine, provisional; decided together with md:120 under C5) and the tree are unchanged. Factual fix: the Bangla bullet originally said the windowed searches returned [] "three times"; the trace shows twice (steps 2 and 3), corrected above.

md:120 · C3 · Multi-domain

- **Task goal:** if average session duration exceeded 19 since Nov 27, create the backlog task 'Improve average session duration' on Front end for the person with the most completed front-end tasks, due **next Friday**. Ground truth: `due_date=2023-12-08`.
- **English:** checked the analytics and completed tasks (nia), then `create_task(..., due_date=2023-12-08)`. Pass.
- **Bangla:** "... ডেডলাইন আগামী শুক্রবার". The analytics, the assignee and the call are identical except `due_date=2023-12-01`. Its answer: "আগামী শুক্রবার (১ ডিসেম্বর, ২০২৩)".
- **First divergence:** step 2, the due_date.
- **Final failed / missing action:** `create_task(..., due_date=2023-12-01)`.
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Root cause:** "আগামী শুক্রবার" was read as "the coming Friday", which is tomorrow on a Thursday. The benchmark means the Friday of next week (12-08).
- **Confound:** Translation/localization error (C3)
- **Evidence:** In multi_domain:120, English passed by calling `create_task` with `due_date=2023-12-08`. In Bangla, the agent read "আগামী শুক্রবার" as 1 December and called `create_task` with `due_date=2023-12-01`. This is labelled a slot-grounding failure caused by a translation/localization ambiguity (provisional), not a general weakness, because no C0 analogue exists.
- **Adjudication:** the failure type is now **Understanding failure**, because the type follows the first failure point (C1). Translation stays the confound. Decided together with email:41 under C5:

md:84 · C6 · Multi-domain

- **Task goal:** if olga has overdue tasks, book a 30-minute 'Catch up on overdue tasks' meeting with her at the earliest free time tomorrow. Ground truth: 2023-12-01 13:00.
- **English:** found the overdue tasks, searched tomorrow 09:00–18:00, and booked 13:00. Pass.
- **Bangla:** the overdue check, email lookup and calendar search are identical to English. It booked **15:00**, stating "আপনি দুপুর ৩টার পর ফ্রি আছেন" and ignoring the free 13:00–13:30 gap.
- **First divergence:** step 3, the event_start computed from the calendar observation.
- **Final failed / missing action:** `create_event(..., event_start=2023-12-01 15:00:00)`.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** the earliest free slot was computed wrongly from the observed events. The agent treated "free" as "after the last meeting".
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** In multi_domain:84, English passed by calling `create_event` at 13:00. In Bangla, the agent read the same calendar and booked 15:00 ("দুপুর ৩টার পর ফ্রি"), skipping the free 13:00 slot. This is labelled a tool-argument (slot computation) failure and a general model weakness, because C0 multi_domain:102 picks the same wrong 15:00 slot.

Appendix B. Trace notes for the 15 gpt-oss:20b EN-pass/BN-fail cases

Ordered by task ID. Fields follow pipeline section 4. Labels (first failure point, type, confound) are the adjudicated ones from `labels_final.csv`; narrative fields come from `trace_notes.md`.

analytics:105 · C6 · Analytics

- **Task goal:** line plot of the most popular traffic source since November 24 (GT: visits_direct).
- **Bangla:** traffic_source_count for direct and referral, then an empty final answer. Fail.
- **First divergence:** step 2. Bangla stops where English continues to search engine and create_plot.
- **Final failed / missing action:** create_plot was never called.
- **First failure point:** Premature termination · **Type:** Premature abandonment
- **Root cause:** the agent stopped mid-task with an empty answer.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** BN final answer `''` after 2 read-only calls.

analytics:42 · C1 · Analytics

- **Task goal:** bar charts of total visits and engaged users since 2023-11-10.
- **English:** create_plot(2023-11-10..2023-11-30) for total_visits and user_engaged. Pass, since an end of 11-30 is accepted.
- **Bangla:** create_plot(time_min="2023-11-10", time_max="2023-11-10") for both metrics. The final answer says the charts are "২০২৩-১১-১০ তারিখের" (for the date 2023-11-10). Fail.
- **First divergence:** step 0, time_max.
- **Final failed / missing action:** both plots cover a single day.
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Root cause:** "2023-11-10 থেকে" (from/since 2023-11-10) was read as that one day.
- **Confound:** Genuine Bangla-related agent failure (C1)
- **Evidence:** `analytics.create_plot {'time_min': '2023-11-10', 'time_max': '2023-11-10', ...}`.
- **Adjudication:** first failure point Entity or slot grounding (C1: the date phrase was resolved to a different referent, the single day). Type, confound (Genuine, with the analytics:0/:32 caveat) and tree unchanged.

analytics:43 · C6 · Analytics

- **Task goal:** bar charts of engaged users and average session duration since 2023-09-24.
- **English:** two create_plot calls. Pass.
- **Bangla:** errored with `engaged_users_count() got an unexpected keyword argument 'field'`. Trace lost; 1 model call preceded the failure (from the log). The first attempt failed the same way, with `'traffic_source'`.
- **First divergence:** the first tool call.
- **Final failed / missing action:** no plots.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** an invented keyword on an analytics count tool.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** the CSV error string, plus the log.
- **Adjudication:** tree S3 ok (the crashing call names the requested metric, C12) and S7 fail (C3).

analytics:54 · C6 · Analytics

- **Task goal:** if total visits was below 3 at any time in the last week, plot it (GT: no change).
- **English:** total_visits_count(11-23..11-29), lowest 6, so no plot. Pass.
- **Bangla:** the same query. It wrote that visits never fell below 3 but still called create_plot(11-23..11-29, total_visits, line) "as requested". Fail.
- **First divergence:** step 1, an unnecessary create_plot.
- **Final failed / missing action:** the spurious create_plot.
- **First failure point:** Task constraint preservation · **Type:** Wrong tool/action
- **Root cause:** the conditional ("if so") was ignored after being evaluated correctly.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** "কখনও ৩-এর নিচে ছিল না। ... তবে আপনার অনুরোধ অনুযায়ী লাইন চার্ট তৈরি করেছি".
- **Adjudication:** labels kept (Task constraint preservation / Wrong tool/action / General, citing C0 analytics:98 EN); tree S5 na (C3: the only write should not exist). Verification failure was rejected: the agent checked the condition and acted against it.

calendar:20 · C6 · Calendar

- **Task goal:** move my first meeting with kofi on December 4 by 1.5 hours (GT: event 00000109 starts 12:00).
- **English:** search(query "Kofi", 2023-12-04..2023-12-04) → []; search("", same window) → []; widened to 00:00:00–23:59:59 and found 00000109 → update_event(..., "2023-12-04 12:00:00"). Pass.
- **Bangla:** search("kofi", 2023-12-04..2023-12-04) → []; search("kofi@example.com", same window) → []; final answer asked the user for the event ID. Fail.
- **First divergence:** step 1. English widened the window, while Bangla queried an invented address and then gave up.
- **Final failed / missing action:** update_event was never called.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** the agent stopped after empty searches instead of recovering. The zero-width first search is the same in the passing English run, so it is not decisive.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** "দয়া করে ওই মিটিংটির ইভেন্ট আইডি (Event ID) আমাকে জানাবেন কি?"
- **Adjudication:** first failure point Tool-argument construction, failure type Wrong tool argument, tree S5 fail and S7 fail (first fail S5). Under C2 the first failure is the uncorrected zero-width window at step 0, which the passing English run widened; Premature termination needs every step before the stop to be correct. Confound General, now citing C0 project_management:60 and :42 (malformed search, then no recovery).

crm:22 · C6 · CRM

- **Task goal:** add Jordan Moore as a new lead assigned to Akira.
- **English:** find_email_address(Akira) → add_customer(customer_name, assigned_to_email, status="Lead"). Pass.
- **Bangla:** the same lookup, then add_customer with the same three values plus customer_email="", customer_phone="", last_contact_date="", product_interest="", notes="", follow_up_by="". Fail, because empty strings differ from nulls in the final state.
- **First divergence:** step 1, the argument list of add_customer.
- **Final failed / missing action:** add_customer with unrequested empty fields.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** the agent fills optional parameters that the task does not mention.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** `customer_email='', customer_phone='', ..., follow_up_by=''`.

crm:38 · C6 · CRM

- **Task goal:** reassign Taylor Jackson to Raj.
- **English:** search_customers → find_email_address(Raj) → update_customer(00000189, assigned_to_email, raj.patel@atlas.com). Pass.
- **Bangla:** step 0 called the non-existent `crm_search_customers`, which was rejected. The rest is identical to English, so the final DB state is correct, but WorkBench fails any run that contains an unrecognised call. Fail.
- **First divergence:** step 0, tool name.
- **Final failed / missing action:** none in the state. The failure is the scored unrecognised call.
- **First failure point:** Tool selection · **Type:** Wrong tool/action
- **Root cause:** a hallucinated tool name.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** "Tool 'crm_search_customers' not found."

crm:53 · C6 · CRM

- **Task goal:** give Nadia all of Sofia's training customers that are qualified or in proposal (3 updates).
- **English:** two lookups, two filtered searches, three update_customer calls. Pass.
- **Bangla:** errored with `update_customer() got an unexpected keyword argument 'new?'`. Trace lost; 4 model calls preceded the failure (from the log `c6_2026-10-07_05-53-19`; 2 HTTP 500s were retried successfully). The first attempt (`c6_2026-10-05_23-57-37`) failed with the same `'new?'` after 6 calls.
- **First divergence:** the update_customer call, with a malformed keyword.
- **Final failed / missing action:** no update was executed.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** a malformed keyword name (`new?` instead of `new_value`).
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** the CSV error string, plus the log.
- **Adjudication:** tree S7 fail (C3: never `na`; the required writes are missing). S3 stays `na` (C12: the lost trace says nothing about it).

crm:77 · C6 · CRM

- **Task goal:** move customers who haven't responded to a software proposal in 6 weeks to lost (GT: no change; there are none).
- **English:** search_customers(product_interest="Software", status="Proposal", last_contact_date_max="2023-10-19") → [], so it changed nothing. Pass.
- **Bangla:** search_customers(product_interest="Software", last_contact_date_max="2023-10-19"), with no status filter, then set 5 customers with status Won/Qualified/Lead to Lost. Fail.
- **First divergence:** step 0, the status filter is missing.
- **Final failed / missing action:** five spurious update_customer(..., status, "Lost") calls.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** the "proposal" status constraint was dropped.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** `search_customers {'product_interest': 'Software', 'last_contact_date_max': '2023-10-19'}`, followed by 5 updates to Lost.
- **Adjudication:** failure type Understanding failure; confound General model weakness; tree S1 ok · S3 fail · S4 fail · S5 na · S7 ok. The transliteration keeps the word's meaning, and the English 'a proposal' admits the same drop (C9, C5(1)). The same model mapped the same 'প্রপোজালে' correctly in crm:74 BN. A status filter missing from the first search also fails C0 crm:42 EN and md:182 EN. The policy deviation remains flagged for the native-speaker review.

email:51 · C6 · Email

- **Task goal:** forward the latest email about 'Update on Client Appreciation Gala' to santiago.
- **English:** search_emails → find_email_address("Santiago") → forward_email(00000206, santiago.martinez@atlas.com). Pass.
- **Bangla:** search_emails found 00000206, then forward_email(00000206, recipient="santiago@example.com") with no directory lookup. Fail.
- **First divergence:** step 1. Bangla skips `find_email_address` and invents an address.
- **Final failed / missing action:** forward_email to the wrong recipient.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** the recipient name was not grounded to its directory address.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** `email.forward_email {'email_id': '00000206', 'recipient': 'santiago@example.com'}`.
- **Adjudication:** first failure point Tool-argument construction (C8: the address was fabricated, not misread); the failure type, confound and tree are unchanged.

email:66 · C6 · Email

- **Task goal:** forward my most recent email from leila to fatima.
- **English:** search "Leila" → find_email_address("Fatima") → forward_email(00000052, fatima.khan@atlas.com). Pass.
- **Bangla:** search "leila" → forward_email(00000052, "fatima@example.com"). Fail.
- **First divergence:** step 1, no directory lookup.
- **Final failed / missing action:** forward_email to the invented address.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** ungrounded recipient name.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** `recipient='fatima@example.com'`.
- **Adjudication:** first failure point Tool-argument construction (C8: the address was fabricated, not misread); the failure type, confound and tree are unchanged.

email:69 · C6 · Email

- **Task goal:** forward my most recent email from olga to raj.
- **English:** search "Olga" → find_email_address("Raj") → forward_email(00000206, raj.patel@atlas.com). Pass.
- **Bangla:** search "olga" → forward_email(00000206, "raj@example.com"). Fail.
- **First divergence:** step 1, no lookup.
- **Final failed / missing action:** forward_email to the invented address.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** ungrounded recipient name.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** `recipient='raj@example.com'`. The final answer names the wrong person ("রজকে").
- **Adjudication:** first failure point Tool-argument construction (C8: the address was fabricated, not misread); the failure type, confound and tree are unchanged.

email:74 · C6 · Email

- **Task goal:** forward the last 'Update on Team Building Retreat' email to lena and aisha.
- **English:** search → find_email_address(Lena), find_email_address(Aisha) → two forwards of 00000262 to the @atlas.com addresses. Pass.
- **Bangla:** search → forwards of 00000262 to "lena@example.com" and "aisha@example.com". Fail.
- **First divergence:** step 1, no lookups.
- **Final failed / missing action:** both forward_email calls have invented recipients.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** ungrounded recipient names.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** `recipient='lena@example.com'`, `recipient='aisha@example.com'`.
- **Adjudication:** first failure point Tool-argument construction (C8: the address was fabricated, not misread); the failure type, confound and tree are unchanged.

pm:2 · C6 · Project mgmt

- **Task goal:** move all of yuki's in-progress tasks to in review (00000091).
- **English:** find_email_address(Yuki) → search_tasks(yuki.tanaka@atlas.com, In Progress) → update_task(00000091, list_name, "In Review"). Pass.
- **Bangla:** search_tasks("yuki@example.com") → []. find_email_address("yuki") returned yuki.tanaka@atlas.com, but the agent kept searching with "yuki@example.com" (4 empty searches) and concluded there were no tasks. Fail.
- **First divergence:** step 0, the invented assignee address. The lookup result was ignored at step 2.
- **Final failed / missing action:** update_task was never called.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** an invented identifier that persisted despite a correct lookup.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** the obs `["yuki.tanaka@atlas.com"]`, followed by `search_tasks {'assigned_to_email': 'yuki@example.com', ...}`.
- **Adjudication:** first failure point Tool-argument construction (C8: the fabricated address was kept despite the correct lookup); the type, confound and tree are unchanged.

pm:72 · C6 · Project mgmt

- **Task goal:** reassign nia's most urgent task to olga (00000162).
- **English:** lookups for Nia and Olga, search_tasks, then update_task(00000162 → olga.petrova). Pass.
- **Bangla:** 11 search_tasks("nia@example.com") calls, all [], one garbled tool name ("project___...We"), then it asked the user for Nia's task list. Fail.
- **First divergence:** step 0, the invented address with no lookup.
- **Final failed / missing action:** update_task was never called.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Root cause:** an ungrounded assignee identifier, followed by a degenerate retry loop.
- **Confound:** General model weakness present in both English and Bangla (C6)
- **Evidence:** 11× `search_tasks {'assigned_to_email': 'nia@example.com'}` → `[]`.
- **Adjudication:** first failure point Tool-argument construction (C8); the type, confound and tree are unchanged.

Appendix C. EN-fail/BN-pass cases (English side)

gemma4:31b (8 cases)

Task	First failure point (EN)	Failure type	Confound	Evidence
calendar:63	Tool-argument construction	Wrong tool argument	C6	EN failed: booked 'catch-up' at 2023-12-01 11:00, which overlaps the 10:00-12:00 event; ground truth (and BN) is 13:00. Same slot error as C6 multi_domain:84 and multi_domain:200 BN.
crm:42	Task constraint preservation	Dialogue-state failure	C7	EN failed: the search with status='Lead' returned [], so it dropped the lead filter and reassigned 5 non-lead training customers (e.g. 00000186, status Lost) to lena; ground truth is no change, and BN stopped correctly.
crm:46	Tool-argument construction	Wrong tool argument	C7	EN failed: searched Lena's hardware customers without status='Lead', reassigned non-leads 00000165 and 00000066 and missed 00000015; BN filtered status='Lead' and matched the ground truth. ADJ fix round 1 (C1 tie-break): the answer restates 'all of Lena Schmid...
crm:57	Instruction understanding	Understanding failure	C7	EN failed: read 'Give Akira all of Nadia's customers ... in the crm' as emailing Akira a list (send_email) instead of update_customer(assigned_to_email=akira) on 4 customers; BN reassigned all 4 correctly.
md:102	Tool-argument construction	Wrong tool argument	C6	EN failed: booked at 2023-12-01 12:00 (overlaps 12:00-13:00), deleted it and rebooked at 15:00, skipping the free 13:00 slot that ground truth (and BN) uses. Identical 15:00 choice to C6 multi_domain:84 BN.
md:130	Task constraint preservation	Wrong tool/action	C6	EN failed: called create_task('Improve engaged users') at step 3 before evaluating the condition; its final answer then computes -18% vs +150% and says 'the condition was not met ... I have already created the task'. Ground truth is no task, and BN correctly m...
md:6	Tool-argument construction	Wrong tool argument	C6	EN failed: create_event at 2023-12-01 12:00 (overlaps 12:00-13:00), then rebooked at 13:00 and deleted the first event. The final event has ID 00000301 instead of 00000300, so the strict state comparison fails. Same slot error as C6 multi_domain:84 BN.
pm:28	Task constraint preservation	Wrong tool/action	C6	EN failed: moved 00000162 (due 2023-11-30, today) to In Progress, while saying 'No overdue tasks were found'; ground truth is no change, and BN correctly treated due-today as not overdue. Same mechanism as C6 project_management:64 BN.

gpt-oss:20b (10 cases)

Task	First failure point (EN)	Failure type	Confound	Evidence
calendar:105	Tool-argument construction	Wrong tool argument	C6	EN called create_event(participant_email="yuki@example.com", ...) with no lookup (GT yuki.tanaka@atlas.com); same mechanism as C0 workbench:email:21. C7 analogue in C6: workbench:email:51 BN and workbench:project_management:2 BN (yuki@example.com).
calendar:38	Task constraint preservation	Wrong tool/action	C6	EN deleted the next "Quarterly Sales Review" 00000121 (GT) and also 00000178 (2023-08-03, past); deleting past events against a temporal constraint as C0 workbench:calendar:84. C7 analogue in C6: workbench:calendar:84 BN (deleted past meetings against 'সামনে' ...
crm:60	Tool-argument construction	Wrong tool argument	C6	EN searched search_customers(product_interest="Services", status="Won") without Sofia's email and deleted 5 customers of other owners (GT deletes only 00000048); dropped-constraint mechanism as C0 workbench:customer_relationship_manager:42. C7 analogue in C6: ...
crm:74	Tool selection	Wrong tool/action	C6	EN's first call was the non-existent "crm_search_customers" (rejected); it then made the correct search and both GT updates, but the unrecognised call fails WorkBench scoring; same invented name in C6 workbench:customer_relationship_manager:38.
email:21	Tool-argument construction	Wrong tool argument	C6	EN called send_email(recipient="jinsoo@example.com", ...) with no directory lookup (GT jinsoo.kim@atlas.com); same mechanism in C0 workbench:email:75 and C6 email:51.
email:36	Tool-argument construction	Wrong tool argument	C5	The EN task never closes the quote ("with 'Thanks for the update - I will get back to you tomorrow. Can you send the reply for me?"), so EN replied with body "... tomorrow. Can you send the reply for me?" (GT body ends at "tomorrow."); benchmark text defect th...
email:75	Tool-argument construction	Wrong tool argument	C6	EN forwarded 00000120 to "akira@example.com" and "yuki@example.com" without lookups (GT akira.sato, yuki.tanaka); same mechanism as C0 workbench:email:21. C7 analogue in C6: workbench:email:74 BN (the same two-recipient forward to invented addresses) and workb...
pm:28	Task constraint preservation	Wrong tool/action	C6	EN errored: update_task() got an unexpected keyword argument 'new?'; trace lost; 6 model calls preceded the failure (from the log c0_2026-10-05_22-53-07, 5 read timeouts retried). GT is no change, so any update_task call is a write that should not exist. ADJ (...
pm:42	Tool-argument construction	Wrong tool argument	C6	EN searched search_tasks(assigned_to_email="yuki@example.com") 3 times ([]) and concluded there were no tasks (GT update_task 00000091 -> carlos.rodriguez); same as C0 workbench:project_management:60. C7 analogue in C6: workbench:project_management:2 BN (the s...
pm:64	Tool execution	Execution/environment failure	C4	EN errored with "Request timed out." after 9 harness retries; trace lost; 2 model calls preceded the failure (from the log c0_2026-10-05_22-53-07); no model fault visible; GT is no change.

Appendix D. Comparison groups

Confound	gemma PF (BN side)	gemma FF (both sides)	gpt-oss PF (BN side)	gpt-oss FF (both sides)
C1 Genuine Bangla-related agent failure	2	0	1	1
C3 Translation/localization error	1	0	0	2
C4 Tool/environment failure	0	0	0	5
C5 Benchmark/task ambiguity	1	2	0	6
C6 General model weakness present in both languages	3	13	14	39
C7 Unclear	0	3	0	1

Model	Group	Email	Calendar	Analytics	Project mgmt	CRM	Multi-domain	Total
gemma4:31b	EN-pass / BN-pass	14	13	10	13	10	6	66
	EN-pass / BN-fail	1	0	3	0	1	2	7
	EN-fail / BN-pass	0	1	0	1	3	3	8
	EN-fail / BN-fail	0	1	2	1	1	4	9
gpt-oss:20b	EN-pass / BN-pass	7	9	7	6	5	4	38
	EN-pass / BN-fail	4	1	4	2	4	0	15
	EN-fail / BN-pass	3	2	0	3	2	0	10
	EN-fail / BN-fail	1	3	4	4	4	11	27